

Joel Quadras

Full-Stack Software Developer & AI/ML Engineer

Dublin, Ireland • +353 89 437 3029 • joelq.quadras@gmail.com

joel071997.github.io • linkedin.com/in/joelquadras7663 • github.com/joel071997

SUMMARY

Software engineer with 3 years of production experience at Ingram Micro building enterprise data systems, distributed REST and SOAP APIs and CI/CD-driven services for Fortune 500 clients across 140+ countries, plus an MSc in Data Science (Technological University Dublin) and a co-authored peer-reviewed NLP publication. Works across the stack and the full delivery lifecycle: ASP.NET Core and TypeScript services, relational and NoSQL databases, large-scale data pipelines and multi-environment deployment. Recent work is retrieval-augmented generation, with two RAG systems over PostgreSQL tables and document corpora using intent routing, query guardrails and a separated ingestion and serving path. Open to full-stack software and AI/ML engineering roles.

TECHNICAL SKILLS

Languages: Python, SQL, PL/SQL, C#, TypeScript / JavaScript

Distributed Systems & Engineering: Angular, REST / SOAP APIs, ASP.NET Core, microservices, automated pipelines, Azure DevOps, Jenkins, CI/CD, Git, Docker, TDD, multi-environment deployments (Dev / QA / UAT / Staging / Prod)

Data & Warehousing: MS SQL Server, Oracle, PostgreSQL, MongoDB, star-schema data warehouse design, in-database ML, query optimisation, stored procedures, batch and large-scale data processing, SSRS

AI / ML / Generative AI: PyTorch, TensorFlow / Keras, scikit-learn, Hugging Face Transformers, LangChain, LangGraph, Retrieval-Augmented Generation (RAG), agentic patterns, prompt engineering, embeddings, vector databases (FAISS / Chroma), NLP, text classification, text generation

BI & Visualisation: Power BI, Tableau, SSRS, Matplotlib, Seaborn, Plotly

PROFESSIONAL EXPERIENCE

Software Engineer - Ingram Micro SSC Pvt Ltd, Mumbai, India 2019 – 2022

- Built and shipped 15+ REST / SOAP API integrations for Fortune 500 enterprise clients (IBM, Amazon, FedEx, UnitedHealth), automating hardware-lifecycle data flows across 140+ countries and reducing manual operations by 80%.
- Engineered large-scale data pipelines with SQL and .NET Core backend services, enabling real-time tracking of 50,000+ monthly shipments and cutting fulfilment errors by 40%; direct experience with distributed systems and high-throughput data processing.
- Refactored legacy ASP.NET systems and resolved 20+ production defects, improving device-testing module performance by 30% through query optimisation and modular backend architecture.
- Owned the full SDLC across Dev / QA / UAT / Staging environments and authored 50+ test cases (TDD-informed), executed regression and sanity testing, ensured zero-downtime releases, reducing production defects by 35%.
- Delivered the IBM China GARS programme with 0 critical defects and 98% on-time completion; implemented Unicode-compliant Chinese localisation supporting \$2M+ in hardware-lifecycle operations.
- Recognised with 'Excellence' and 'Best Project Delivery' awards plus an IBM client commendation for the China GARS programme.

- Added and updated Angular UI fields on top of the API work in BlueIQ (internal) and CloudBlue (client-facing) to surface new integration data on screen, closing the loop from backend to front end.

PROJECTS

PostgreSQL RAG Chat - Intent-Routed Retrieval over Live Database Tables - Personal Project, GitHub 2026

- Chat interface over live PostgreSQL tables that routes each question by intent: structured asks become read-only SQL returned as a table, semantic asks go through retrieval over chunked rows. The FAISS index re-builds when tables or row counts change.
- Guardrails enforce SELECT-only single statements with a hard LIMIT, and a low retrieval-confidence fallback that declines to answer rather than guessing. Python, LangChain, FAISS, Streamlit, pytest.

Document RAG Pipeline - Ingestion and Serving Split - Personal Project, GitHub 2026

- Two-stage retrieval stack: one notebook loads and chunks source documents into a persisted Chroma vector store, a second assembles the question-answering chain on top.
- Separating ingestion from serving means a corpus can be re-indexed without touching the query path. LangChain, ChromaDB, embeddings, Python.

Accessibility Made Easy - Cloud-Based NLP Service for Dyslexia-Friendly Documents - Group MSc Project, Technological University Dublin (published, EDULEARN24) 2023 – 2024

- Co-developed (4-student team) a cloud-hosted service that reformats Microsoft Word documents for readers with dyslexia, applying evidence-based typography, spacing, layout and colour-contrast rules automatically or under user control.
- Led the NLP feature work including automatic heading generation from term salience, and owned deployment: secure upload and download, privacy-first architecture and runtime.
- Processed 500+ document conversions with 95% positive user feedback; co-authored peer-reviewed publication at EDULEARN24 (IATED, Palma, July 2024). DOI: 10.21125/edulearn.2024.1392.

Fake News Detection with Explainable AI - Personal Project, GitHub 2024

- Fine-tuned BERT classifier wrapped in LIME so every prediction returns the tokens that drove it, making outputs auditable by a domain expert rather than a single accuracy number. PyTorch, Transformers, Jupyter.

EDUCATION

MSc Data Science - Technological University Dublin 2022 – 2024

Grade: 2.1. Coursework: Machine Learning, Deep Learning, Natural Language Processing, Text Mining, Big Data Analytics, Data Warehousing & In-Database ML, Statistical Methods, Data Visualisation.

BSc Computer Science - Fr. Conceicao Rodrigues College of Engineering 2015 – 2019 Mumbai, India. Grade: First Class.

ADDITIONAL EXPERIENCE

Supervisor — Centz Retail Holding Limited, Dublin 2023 – 2026

Built a small in-store delivery and stock-tracking application used daily by floor staff, improving inventory accuracy. Promoted to Supervisor within nine months, running floor operations alongside MSc studies.